



Checkpoint IV: Second Prototype

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Date: 2025/10/05

Prototype Architecture

The architecture was extended from a single-idiom prototype into a multi-idiom, preprocessed dashboard, with emphasis on efficiency and coordinated interaction. Key changes are:

1. **Global State**
 - Selection now uses a `selectedCards` array (instead of a single `selectedCard` object), enabling multi-card comparisons.
 - `updateSelectionAcrossPlots()` was generalized to handle scatterplot circles, bar chart bars, and line chart paths, ensuring consistent highlighting across idioms.
2. **Preprocessing Pipeline**
 - Data for scatterplot, bar chart and line chart is now pre-aggregated at load time (`scatterTable` (reused for the bar chart) and `lineTable`).
 - This avoids recomputing metrics (totals, averages, condition normalization) after every filter change, significantly improving responsiveness.
 - Filters now only slice the pre-processed tables rather than recalculate from raw CSV.
3. **Expanded Idiom Modules**
 - **Bar chart** and **line chart** are fully implemented and integrated alongside the scatterplot.
 - Each chart has its own module (`createBarchart`, `createLineChart`, etc.), keeping the system modular and extensible.
 - `updateCharts()` orchestrates updates across all idioms using the same filtered subset of data.
4. **Integration Mechanism**
 - Linked highlighting is now bidirectional: selections in scatterplot, bar chart, or line chart propagate across all views.
 - Global deselection (click outside elements) resets the state consistently in all charts.
5. **Usability Enhancements**
 - Added filter count indicators, “select all” logic with lock to prevent rapid refreshes, and a search bar inside filter groups.
 - New tooltip/info icon provides interaction hints, improving user onboarding.

Dashboard Layout

The overall layout (left-hand filters and right-hand visualization grid) remains unchanged from CP III, but two refinements were added:

1. **Year Filters** – years are now displayed in parallel columns, reducing scrolling and empty space.
2. **Autocomplete Search** – filters with many options (e.g., sets, types) now include search bars with autocomplete, helping users quickly locate specific filters.
3. **Information Icon** – a help icon with tooltip was added to provide users with interaction hints.

Aside from these improvements, the chart areas remain the same, with scatterplot and bar chart on the top row, line chart and violin plot placeholder on the bottom row.

Data Processing

In this checkpoint, we expanded preprocessing to support two new idioms:

- Line Chart which answers the Task 3: How does condition affect price in relation to set, type and year?
- Bar Chart which answers the Task 1: Are there any favourite or more popular cards per set and per type?

The biggest improvement since CP III is the introduction of **preprocessing pipelines** at data load, which avoids recalculating metrics after each filter change and makes interactions more responsive.

1. Scatterplot & Bar Chart

- Data is pre-aggregated with d3.rollup into a scatterTable.
- For each card (name–series–set), total sales volume, total revenue, type, and release year are stored.
- The **bar chart** reuses this table and applies an additional aggregation and ranking step to extract the **Top 10 most sold cards** for visualization.

2. Line Chart

- A second table (lineTable) is precomputed, normalizing card **conditions** (mint → poor) and storing average prices per condition.
- Values are sorted in condition order, producing consistent multi-line arrays ready for plotting.

3. Efficiency Gains

- Unlike CP III, where metrics were recomputed after each filter change, CP IV builds **scatterTable** and **lineTable** once during data load. At runtime filtering now only slices these pre-aggregated tables, ensuring faster updates and smoother interactivity.
- This ensures fast updates and smooth transitions even when multiple idioms are active.

Chart Interaction

The scatterplot interactions from CP III (hover tooltips, click-to-select, global deselect, opacity-based highlighting) are maintained.

New in CP IV:

- **Bar Chart:** Hover shows detailed tooltips (card, set, type, total sales). Clicking a bar selects that card, with Ctrl/Cmd+click allowing multi-selection.
- **Line Chart:** Hover over lines or points reveals condition-specific prices. Clicking selects the corresponding card, supporting both single and multiple selections. A zoom/slider interaction was added to adjust the y-axis for detailed analysis of price differences across conditions.

These additions extend the interactive vocabulary of the dashboard while preserving consistent behavior (same selection model across idioms).

Chart Integration

The integration model from CP III (shared data, linked views, global deselect) is maintained, but extended to cover the new idioms.

New in CP IV:

- **Bidirectional Linking:** Selecting a card in the bar chart or line chart now updates the scatterplot and vice versa.
- **Multi-selection Support:** Users can now select multiple cards simultaneously, with all selected items highlighted across views.

The mechanism remains modular: each chart listens to the same global selectedCards array, ensuring that future idioms (e.g., violin plot) will integrate without structural changes.